

Solution Code - Microservices - Composite_Microservices

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: src\main\java\com\example\microservices\CompositeApplication.java

```
1 | package com.example.microservices;
2 |
3 | import org.springframework.boot.SpringApplication;
4 | import org.springframework.boot.autoconfigure.SpringBootApplication;
5 | import org.springframework.web.bind.annotation.*;
6 | import org.springframework.web.client.RestTemplate;
7 |
8 | @RestController
9 | class CompositeController {
10 |     private RestTemplate restTemplate = new RestTemplate();
11 |
12 |     @GetMapping("/portal/student/{id}")
13 |     public String getPortalStudentInfo(@PathVariable String id) {
14 |         String student = restTemplate.getForObject("http://localhost:8081/students/" + id, String.class);
15 |         String courses = restTemplate.getForObject("http://localhost:8082/courses/student/" + id,
String.class);
16 |         return "{ \"student\": " + student + ", \"courses\": " + courses + " }";
17 |     }
18 | }
19 |
20 | @SpringBootApplication
21 | public class CompositeApplication {
22 |     public static void main(String[] args) {
23 |         SpringApplication.run(CompositeApplication.class, args);
24 |     }
25 | }
```

Source File: src\main\java\com\example\microservices\ServiceApplication.java

```
1 | package com.example.microservices;
2 |
3 | import org.springframework.boot.SpringApplication;
4 | import org.springframework.boot.autoconfigure.SpringBootApplication;
5 | import org.springframework.cloud.client.discovery.EnableDiscoveryClient;
6 |
7 | @EnableDiscoveryClient
8 | @SpringBootApplication
9 | public class ServiceApplication {
10 |     public static void main(String[] args) {
11 |         SpringApplication.run(ServiceApplication.class, args);
12 |         System.out.println("Microservice Exercise 5 running...");
13 |     }
14 | }
```